Rijksuniversiteit Groningen

Master's Thesis Economics

Data and Methodology Outline

Barnabás Zoltán Bognár

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Supervisors: Y. Xu, PhD
dr. J.M. Berk

1 Preface

Please note that due to time constraints the literature review has been omitted and the sections are intentionally brief. The data section is still heavily a work in progress as some data has proven to be especially tricky to obtain from the ECB.

2 Methodology

To conduct our analysis we broadly use the methodology of Ricco et al. (2024). The authors use an IV approach to estimate the effects of normal monetary policy, forward guidance, QE and country specific factors on the market. The authors report that their methods resolve market puzzles during periods of acute market stress by incorporating the nonlinear effects of information. We will largely follow the authors' methodology, but introduce the key difference of including the holdings of sovereign debt by banks which allows us not only to observe indicators of the doom loop, but direct evidence for it.

To begin, we construct a duration-adjusted exposure index defined as

$$E_{c,t} = \frac{D_{c,t}^{mod} \cdot H_{c,d}}{K_{c,t}} \cdot \Delta s_{c,t} \tag{1}$$

where

$$D_{c,t}^{mod} = \frac{D_{c,t}^{Macaulay}}{1 + y_{c,t}} \text{ and } D_{c,t}^{Macaulay} = \frac{\sum_{i \in c} \sum_{k=1}^{\max\{m\}} m_k \cdot H_{i,c,t,k}}{\sum_{i \in c} \sum_{k=1}^{\max\{m\}} \cdot H_{i,c,t,k}}$$

and $E_{c,t}$ is the index, $D_{c,t}^{mod}$ is the modified duration of the bond portfolio, $H_{c,t}$ is the notational carrying amount of domestic sovereign debt securities held, $K_{c,t}$ is Tier 1 capital held, $\Delta s_{c,t}$ is the change in the sovereign spread compared to baseline (in our case, the baseline is Germany), $y_{c,t}$ is the prevailing sovereign yield, while m_k is the maturity of a bond. Here c signifies country, t signifies times, i signifies bank and k signifies maturity bucket. Note that $H_{c,t}$ and $H_{i,c,t,k}$ differ. For the former, aggregate bond holdings of all financial institutions in country c are used, while for the latter, only the banks that data is consistently available for are used. While these two measures are intrinsically different, the banks with available data tend to be large and therefore their duration estimates will tend to be representative. Nevertheless, we note that the Macaulay duration can be approximated as

$$D_{c,t}^{Macaulay} \approx D_t^{10Y} = \frac{1 - (1 + y_{c,t})^{-10}}{y_{c,t}} + \frac{10}{(1 + y_{c,t})^{10}}$$

or the closed-form duration of a ten-year bullet bond with coupon equal to its yield to maturity. As a robustness check, we will employ this specification as well. Another notable element is the classification of maturity. The EBA Transparency Exercise we

employ contains 7 maturities (as such $\max\{m\} = 7$) which represent the following actual maturities.

Table 1: EBA Transparency Exercise Maturity Nomenclature

Bucket	Maturity Range	Midpoint (m_k)
1	0-3 Months	0.125
2	3-12 Months	0.625
3	1-2 Years	1.5
4	2-3 Years	2.5
5	3-5 Years	4.0
6	5-10 Years	7.5
7	10+ Years	15

The construction of the index serves to control for the diverging movements in bond prices due to interest rate adjustments. When interest rates increase bonds with shorter maturities are relatively less affected than bonds with longer maturities. This is important to note as, while certain literature (Albertazzi et al., 2021) finds no evidence of QE inducing term-preference changes for banks, our evidence as presented in Appendix A provides anecdotal evidence that QT may have nonetheless had an effect. Glancing at Figures 13, 14, 15, 17, 18, 20 and 22 we see that for 7 out of our 10 analysed countries there is a clear shift towards longer maturities. This is seemingly at odds with the conclusion of the literature that QE did not cause maturity transformation (as such we wouldn't expect QT to have an effect either). It would thus seem that banks have now begun searching for yield through maturity.

To continue, we adapt the approach Ricco et al. (2024), specifically their Equation 24

to run the following non-liner information regression.

$$
\begin{aligned}
ms_t^i =& \beta_0 + \sum_{j=0}^{J} \theta_j F_t x_{q+j} + \sum_{j=0}^{J-1} \eta_j \Delta F_t x_{q+j} \\
&+ I(S_t > \overline{s}) \left[\sum_{j=0}^{J} \kappa_j F_t x_{q+j} + \sum_{j=0}^{J-1} \psi_j \Delta F_t x_{q+j} \right] \\
&+ I(QT = 1) \left[\sum_{j=0}^{J} \lambda_j F_t x_{q+j} + \sum_{j=0}^{J-1} \mu_j \Delta F_t x_{q+j} \right] \\
&+ I(E_{c,t} > \overline{e}) \left[\sum_{j=0}^{J} \phi_j F_t x_{q+j} + \sum_{j=0}^{J-1} \rho_j \Delta F_t x_{q+j} \right] \\
&+ I(S_t > \overline{s}) \times I(QT = 1) \left[\sum_{j=0}^{J} \alpha_j F_t x_{q+j} + \sum_{j=0}^{J-1} \xi_j \Delta F_t x_{q+j} \right] \\
&+ I(E_{c,t} > \overline{e}) \times I(QT = 1) \left[\sum_{j=0}^{J} \gamma_j F_t x_{q+j} + \sum_{j=0}^{J-1} \delta_j \Delta F_t x_{q+j} \right] \\
&+ I(S_t > \overline{s}) \times I(E_{c,t} > \overline{e}) \times I(QT = 1) \left[\sum_{j=0}^{J} \sigma_j F_t x_{q+j} + \sum_{j=0}^{J-1} \tau_j \Delta F_t x_{q+j} \right] \\
&+ \widetilde{\widetilde{ms_t^i}}
\end{aligned}
\tag{2}
$$

The notation here requires quite a rigorous explanation. First, ms_t^i denotes the market price revision of asset i at time j. Here we use OIS spreads at different maturities, sovereign spreads, exchange rates and STOXX50 stock surprises. Naturally, this requires the rerunning of the regression several times. Second, $F_t x_{q+j}$ denote the level forecast of variable x at horizon $q+j$, while $\Delta F_t x_{q+j}$ denote the revision of the same. The variable x here is effectively a vector of three different variables, namely GDP, HICP and MRO both from the ECB projections as well as the Reuters pre-meeting polls, creating a vector of size six. Notably, the access to the Reuters data is unclear for now, in its absence, the ECB's quarterly Survey of Professional Forecasters may be employed (a method endorsed as an alternative by Ricco et al. (2024) themselves). As this data is quarterly while the ECB meetings occur eight times annually, the last available value will be taken as given at each meeting should it be used.

The horizon J is not clearly restricted by the authors, but its size is set to be 2, implying a look-ahead of two periods both in the case of quarterly and annual projections. This limit is sensible as longer term forecasts have relatively little bearing on asset prices. Next, $I(S_t > \overline{s})$ indicates times of acute market stress when the VSTOXX index is more than one standard deviation outside its mean, $I(QT = 1)$ indicates whether QT took place in a period, while $I(E_{c,t} > \overline{e})$ indicates periods when bank exposure is more than

one standard deviation outside of its mean. Finally, and perhaps most importantly, $\widetilde{\widetilde{ms}}_t^i$, the residual is the informationally robust monetary policy surprise which has been purged of all other factors. This equation then allows us to isolate the effects on the market by unexpected ECB announcements while specifically allowing these effects to vary between times of QT and baseline regimes. However, these market surprises are an aggregate and do not allow us to instrument the different factors impacting the economy. Ricco et al. (2024) use a complex set of mathematics to isolate these effects which can be found in Appendix B.1. The authors thus arrive at four factors as in Table 2.

Table 2: Instruments and Their Interpretation

Factor	Source
F1	ECB announcement of current policy rate
F2	ECB announcement of future interest rates
F3	ECB announcement of QE/QT including APP and PEPP
F4	ECB announcement of peripheral policy

Naturally, of the four factors, F4 is of the most relevance to our research. It represents the surprise announcements of the ECB concerning peripheral countries that then affected spreads. Such announcements could include instances where the market observed that the Bank was unable or unwilling to stabilize the spread between core and peripheral sovereign debts, or, on the contrary, that the ECB was willing to do 'whatever it takes' as enunciated by Draghi (2012).

Having obtained our instruments, our analysis diverges in two separate directions. The first follows the Bayesian Vector Autoregression (BVAR) methodology of Ricco et al. (2024). A base VAR specification takes the form

$$y_t = c + \sum_{\ell=1}^{p} B_\ell y_{t-\ell} + u_t \tag{3}$$

where y_t is the $n \times 1$ vector of endogenous variables, c is a similar sized vector of constants, and B_ℓ is the $n \times n$ matrix of autoregressive coefficients at lag ℓ. Following the methodology of Ricco et al. (2024), we set $p = 12$. Finally, u_t is the vector of residuals with covariance matrix $\mathbb{E}[u_t, u_t'] = \Sigma_u$. The authors establish a baseline specification with their VAR. Limiting our approach to peripheral countries and adapting the specification for our data, this would yield

$$y_t = \begin{pmatrix} \text{IPC}_t \\ \text{GDP}_t \\ \text{HICP}_t \\ \text{CoreEA}_t \\ \text{GDP Deflator}_t \\ \text{OIS}_t^{1m} \\ \text{OIS}_t^{3m} \\ \text{OIS}_t^{1y} \\ \text{OIS}_t^{2y} \\ \text{OIS}_t^{10y} \\ \text{s}_t^{ITA-GER} \\ \text{CDS}_t^{bank} \\ \text{E}_t \\ \text{TARGET 2}_t \\ \text{EUR/USD}_t \end{pmatrix}$$

where all variables except rates are in log-levels. However, this specification does not allow for a distinction between separate peripheral countries. Their inclusion, on the other hand, would require us to estimate roughly $32^2 \times 12 = 12,288$ autoregressive coefficients with approximately 132 monthly observations. Such an estimation would be profoundly inefficient, as such, we will run a baseline specification, as described above, granting us insight into the core-periphery dynamic, and will thereafter rerun the regression five times, once for each member of GIIPS, replacing s_t, CDS_t, E_t and TARGET 2_t with country-specific measures.

Note, however, that our VAR still struggles with its degrees of freedom. With 16 variables across 12 lags the number of variables becomes greater than 3200, or roughly 24 parameters for each observation. This would produce completely unreliable estimates, overfit the data and produce utterly useless out-of-sample estimates. The solution proposed by the authors, and our choice, is the Bayesian approach, which incorporates prior beliefs about the likely values of variables before relying on the data itself. Formally this takes the form

$$p(B, \Sigma_u|Y) \propto p(Y|B, \Sigma_u) \cdot p(B, \Sigma_u)$$

where $p(B, \Sigma_u|Y)$ is the posterior, or the belief after observation, $p(Y|B, \Sigma_u)$ is the likelihood, or the probability of the data given the parameters, and $p(B, \Sigma_u)$ is the prior, or the belief before observation. Of key importance is the prior, which allows us to ground our beliefs in sound economic expectations instead of wildly oscillating upon market noise. Ricco et al. (2024) then make an important assumption, namely that the

priors approximate a random walk, as introduced by Litterman (1981), who observe that most macroeconomic variables tend to follow such processes. These priors are otherwise known as Minnesota priors after the university where the author obtained their PhD. Random walks (without drift) are described by

$$Y_t = Y_{t-1} + \varepsilon_t$$

with $\mathbb{E}[\varepsilon_t] = 0$ resulting in a unit root. In economic terms this implies that the forward-looking best-guess about any economic variable is its current value. Formally, we take B to be

$$\mathbb{E}[B_{\ell,ij}] = \begin{cases} \delta_i = 1 & \text{for } i = j \text{ and } \ell = 1 \\ 0 & \text{otherwise} \end{cases}.$$

This specification implies that we expect the variables to be affected by their first lag and their first lag only, while expectations attached to later lags and cross-variable effects are reverted to their mean. Notably this does not imply that we reject cross-variable effects, but that our posterior beliefs must be heavily influenced by the data to be affected. In addition to the specification of the mean, we also specify the variance, or our confidence that the mean is correct. The Minnesota priors follow the specification

$$\mathrm{Var}(B_{\ell,ij}) = \begin{cases} \frac{\lambda^2}{\ell^2} & \text{if } i = j \\ \frac{\lambda^2 \sigma_i^2}{\lambda^2 \sigma_j^2} & \text{if } i \neq j \end{cases}$$

where λ represents a tightness parameter. The smaller its value, the more the posterior estimate is pulled towards a random walk, while the standard OLS/VAR specification reemerges as $\lambda \to \infty$. While this magnitude is often assumed, employing the work of Giannone et al. (2015), we treat λ as an unknown hyperparameter and estimate it by maximising the marginal likelihood of the data (mathematics in Appendix B.2). The parameters σ_i and σ_j are added to control for the difference in variance between variables and are estimated using an AR(p) process prior to the VAR.

Altogether, the BVAR approach allows us to estimate a VAR model with relatively few observations and many parameters as it stabilizes the estimates even in the face of market noise, while also allowing for the inclusion of variables, such as $E_{c,t}$ which occur at a quarterly and not monthly frequency, and with which the general VAR struggles. This is done through a Gibbs Sampler estimation (mathematics in Appendix B.3), a process which allows for the sampling of the conditional distribution of each parameter block given the others and the subsequent estimation of the posterior without actual observation. Given sufficient iterations, estimates converge to true posterior values, solving the data constraints.

3 Data

To begin this section we turn to the discussion of the literature found in Gómez-Puig and Sosvilla-Rivero (2024). The authors highlight the fundamental condition of the doom loop, that banks have a persistent home bias, or the sovereign-bank nexus as named in the paper. While this bias has been widely documented, we nonetheless turn to the EBA Transparency Exercise to retrieve our first chunk of data on the sovereign bond holdings of Euro Area banks including their maturities and the issuing countries. Our data spans from 2018 December to 2025 June. It is worthwhile to note that earlier data are not cleanly provided by the EBA as issuing countries cannot be identified, while later data does not currently exist. The data is semi-annual with one annual gap between 2019 June and 2020 June due to the COVID pandemic.

To make our data appropriate for analysis, we exclude Croatian and Bulgarian entities as the countries have only recently joined the EA, while also excluding the non-EA bond holdings of remaining banks, as well as those banks not observed in every time period. In our final dataset we thus include 64 separate entities from 15 countries (out of a possible 19) with all 5 GIIPS countries being represented within our sample. To review these holdings, visualisation is available in Appendix A. The overarching conclusion is clear; home bias is persistent for all core and periphery countries analysed. A few curious tendencies can be observed from the data. First, German Bunds are only the most popular sovereign debt holding for Greek and Italian banks (Figures 6 and 8), with its popularity exploding for the Greeks post-2022 and French government bonds taking over in the case of the latter from mid-2023. For the rest of the GIIPS countries French holdings are significantly more popular. German and French banks also seemingly prefer to hold Italian bonds (Figures 4 and 5) with the rest of the core rather holding Spanish bonds. Notably Belgian and Austrian banks terminate their Greek bond holdings at the start of the Covid pandemic, with Dutch banks following suit mid-2024.

A great deal of literature has been devoted to examining the emergence of the doom loop and its possible quantifiers. An approach which is widely used by, among others, Angeloni et al. (2024), Salvatore et al. (2022), and Yu (2017), uses sovereign and bank CDS spreads as well as co-movements to infer the existence of the doom loop. Unfortunately this is not public data and only LSEG contains data for it. While this has not yet been obtained, the request will be handed in this week at the latest. A key component of the doom loop is the TARGET 2 imbalance. The TARGET mechanism is used inside of the EA to settle cross-border payments between financial institutions. When a bank in country, for instance, Italy, transfers money to a German counterparty in the form of reserves, the counterparty reserves this balance immediately. However, the balance sheet of the respective NCBs must also adjust to reflect this change. Namely, reserves flow from the liabilities of the Italian central bank to the Bundesbank. However, this would

result in excessively large assets for the Italian and liabilities for the German bank. As such both central banks create a TARGET balance on their sheets in order to balance their books. Notably the TARGET balances are settled dynamically so that inverse flows negate each other and balances therefore reflect final amounts and not aggregate inflows. Countries that hold positive amounts of TARGET balances are known as creditors, while their counterparts are known as debtors. These balances are somewhat contentious and may pose significant risk (Blake, 2018; Cochrane et al., 2025; Horst & Neyer, 2022; Menrad, 2020; Minenna et al., 2018). For the sake of brevity (and lack of time) it is notable that the most cited concern is the (partial) dissolution of the EU. TARGET balances can effectively be viewed as off-balance-sheet uncollateralized debt, the settlement of which upon exit is unclear in the current TARGET 2 framework.

These (im)balances have increased during the period of QE. To confirm this we may consult Figure 11, which shows an explosion of German TARGET balances after 2014 with a corresponding nosedive of Italian and Spanish balances. The precise reason behind this phenomenon is unclear. Horst and Neyer (2022) and Menrad (2020) conclude that QE inevitably results in this divergence due to the nature of the APP with NCBs purchasing their own government bonds mostly from foreign investors who hold accounts in major financial centres such as Germany. Minenna et al. (2018) offer a contrarian view and argue that QE has effectively enabled peripheral investors to shift their security holdings to core country assets, while the anomalous German balance was driven by a persistent balance of payment surplus and a devalued euro.

In either case, however, QT is expected to shrink these imbalances through one of the channels. To the extent that this does not happen, this divergence could signal capital flight from the periphery towards the core despite the counteracting effect of normalisation. Indeed, this is what the graphs show. The TARGET balance of Germany, Italy and Spain stagnate. Notably, this persistent imbalance may be a result of a multitude of factors which will be controlled for during further analysis.

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A Figures

Figure 1: Austrian Bond Holdings

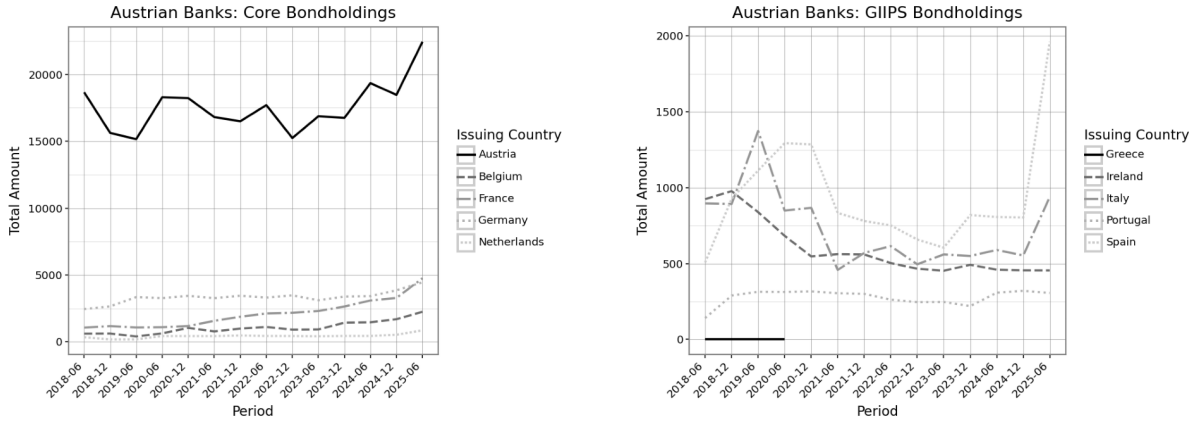


Figure 2: Belgian Bond Holdings

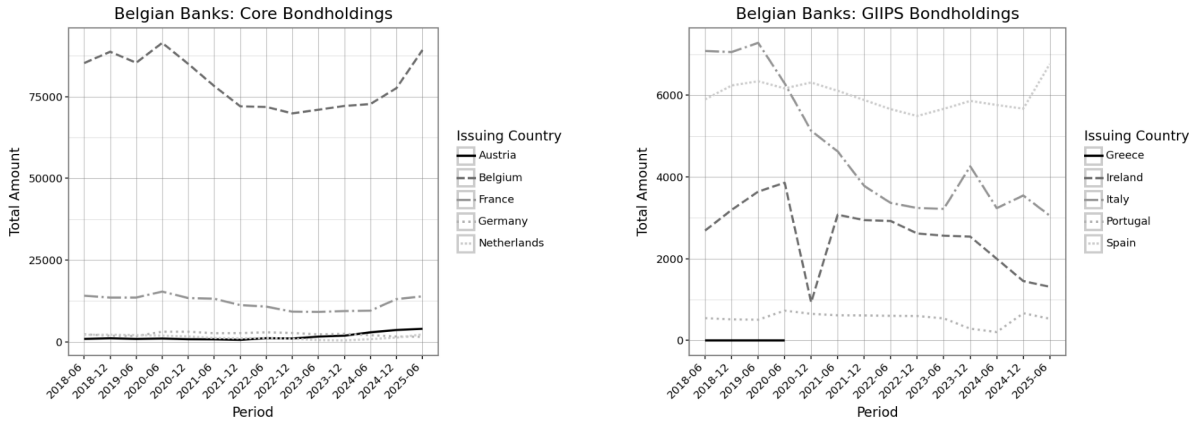


Figure 3: Dutch Bond Holdings

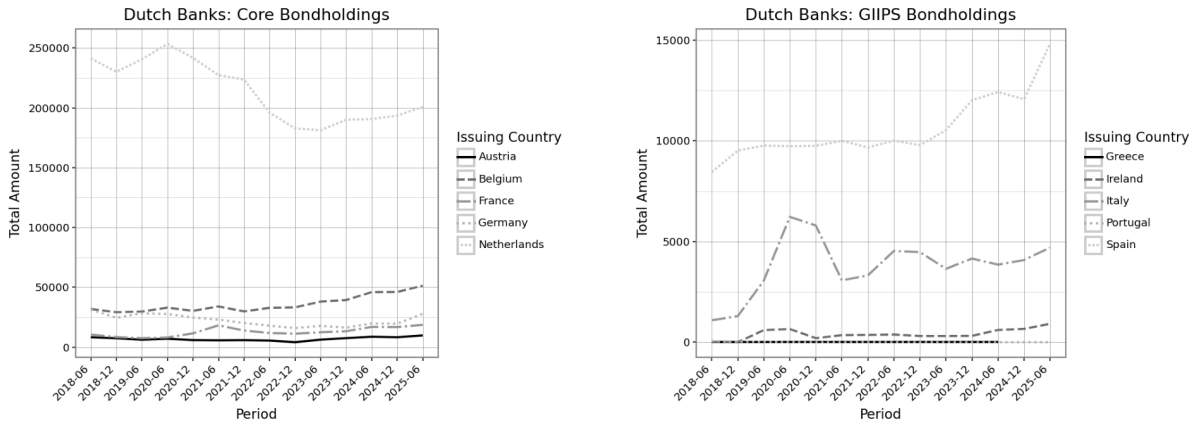


Figure 4: French Bond Holdings

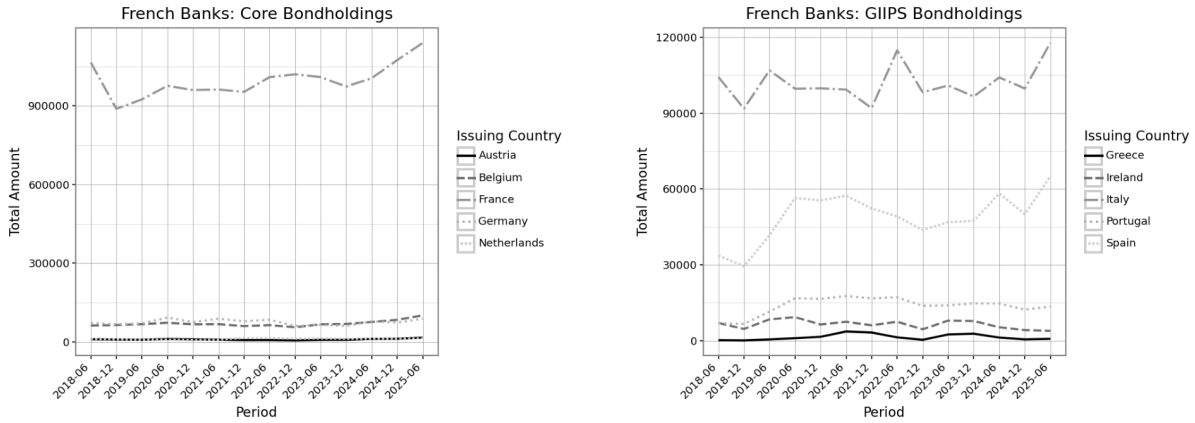


Figure 5: German Bond Holdings

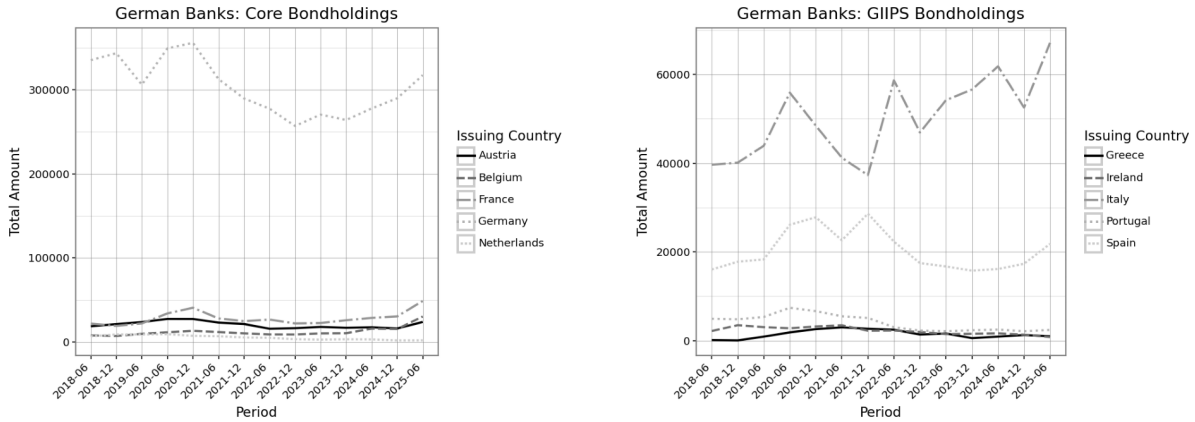


Figure 6: Greek Bond Holdings

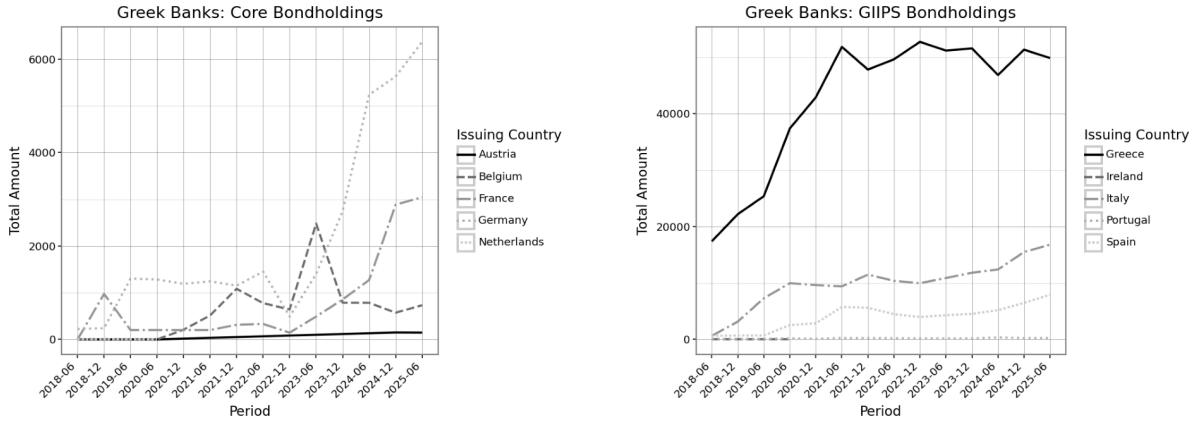


Figure 7: Irish Bond Holdings

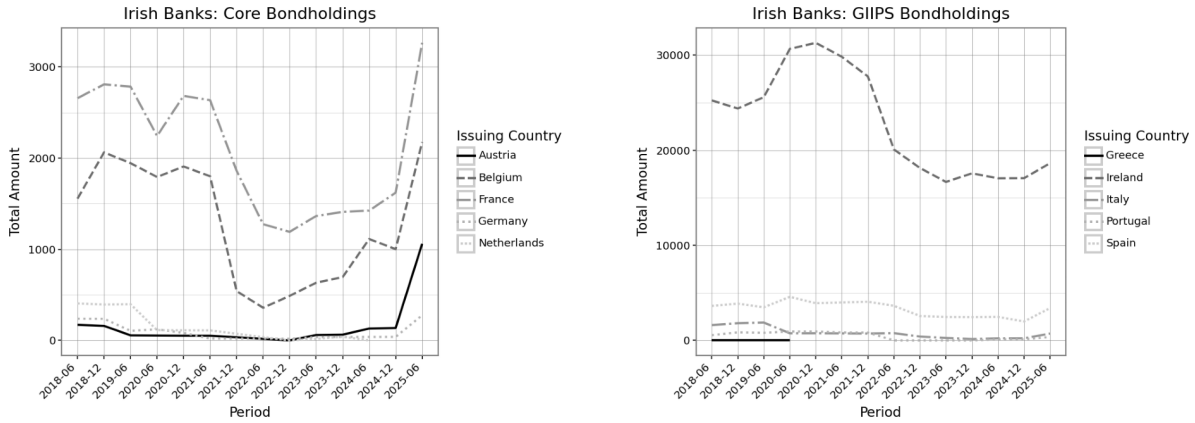


Figure 8: Italian Bond Holdings

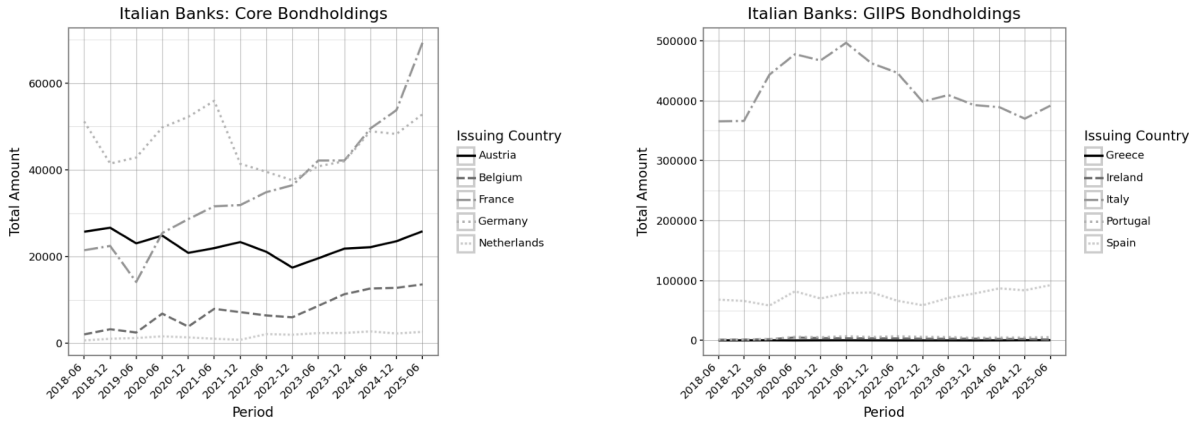


Figure 9: Portuguese Bond Holdings

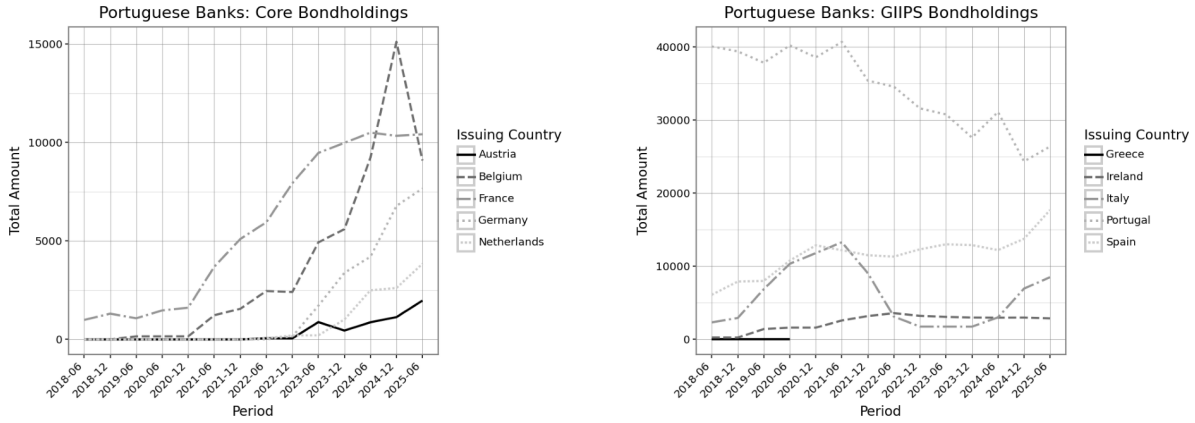


Figure 10: Spanish Bond Holdings

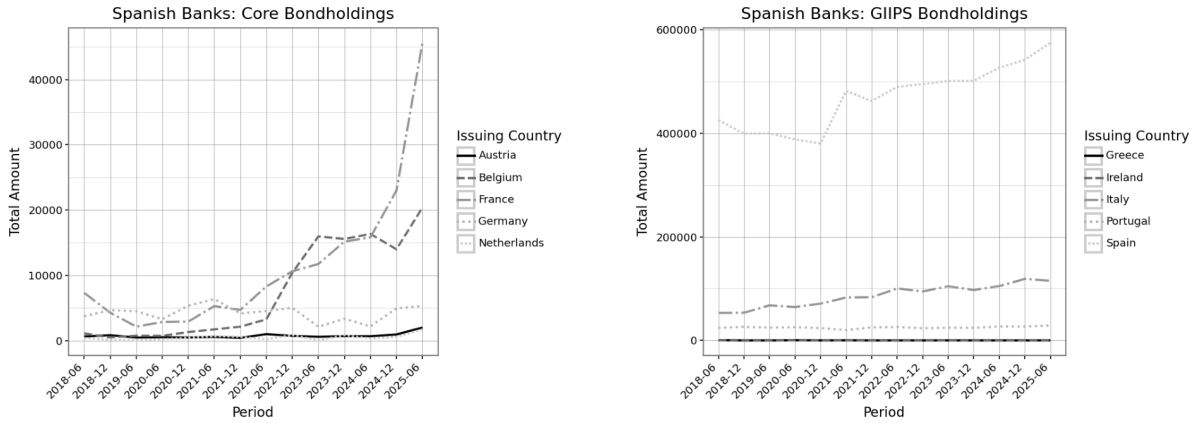


Figure 11: TARGET Balances Since 2008

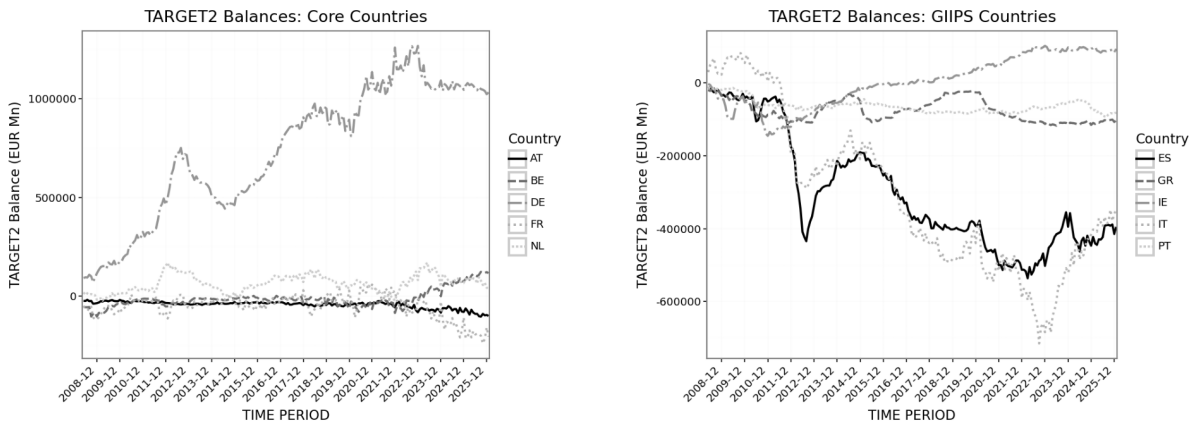


Figure 12: TARGET Balances Since 2018

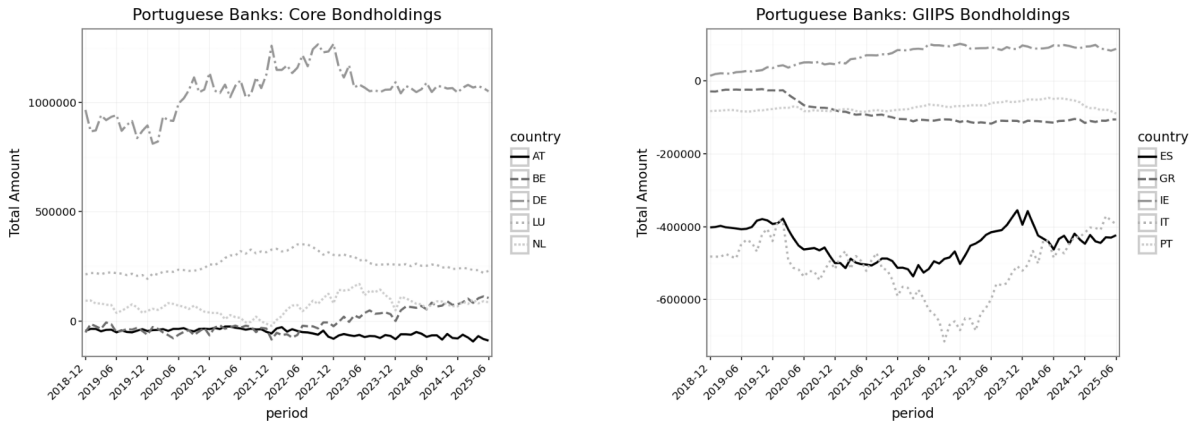


Figure 13: Austrian Bond Holdings by Maturity

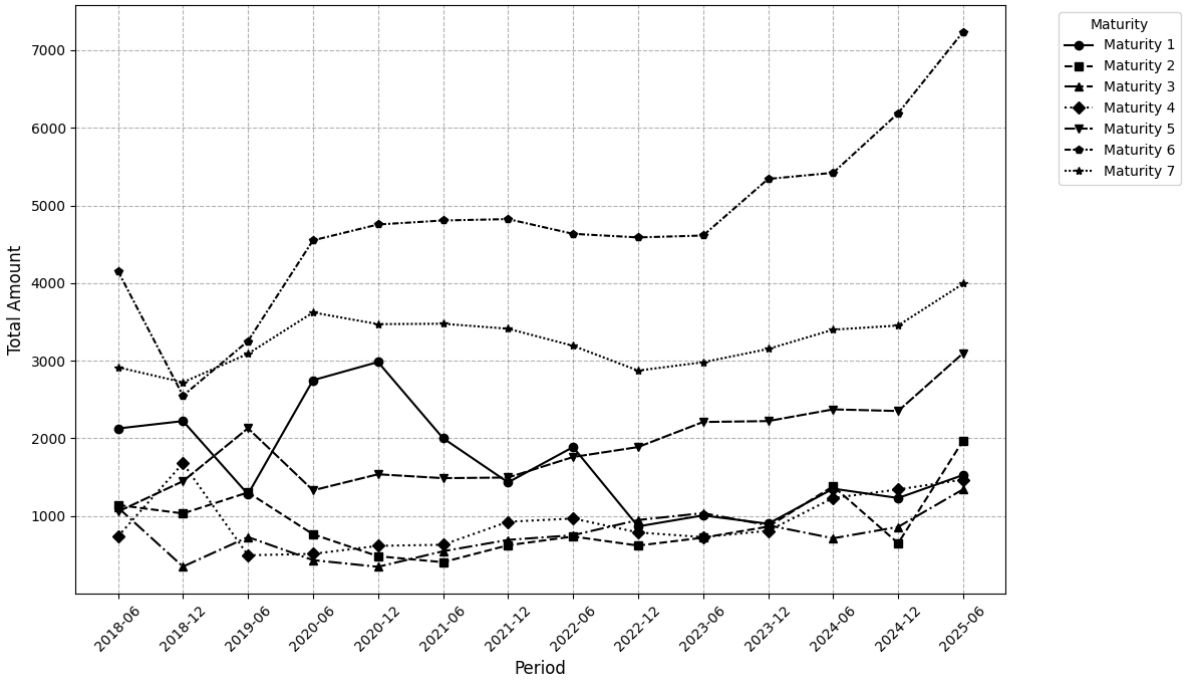


Figure 14: Belgian Bond Holdings by Maturity

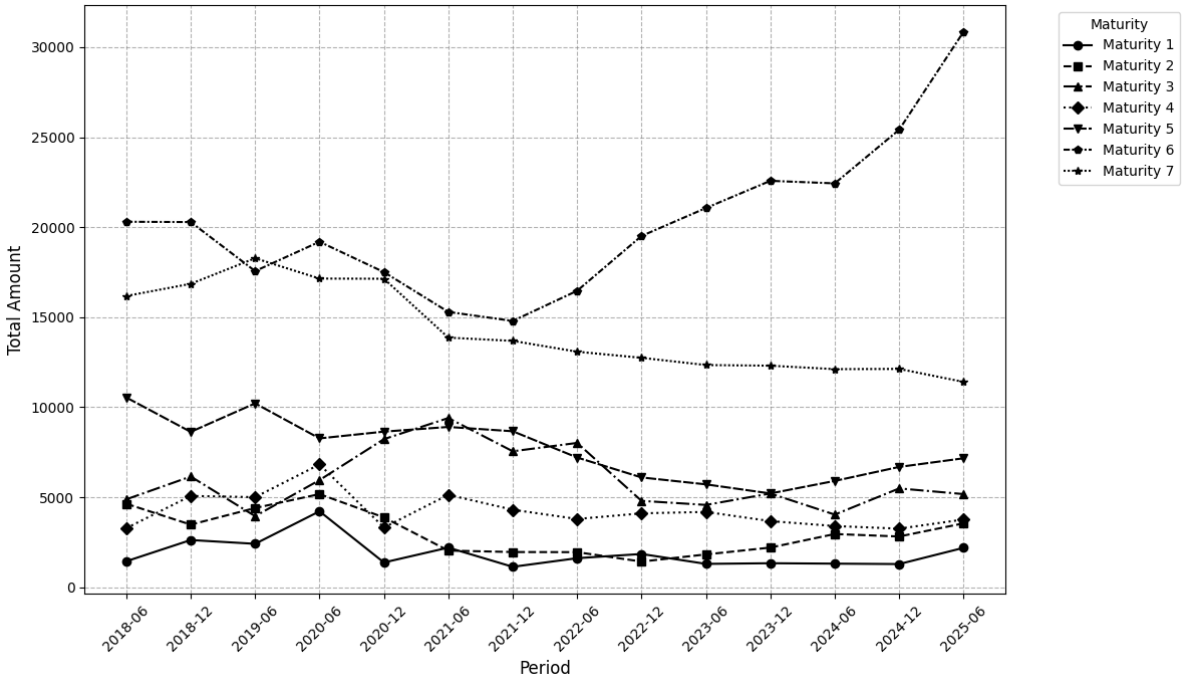


Figure 15: Dutch Bond Holdings by Maturity

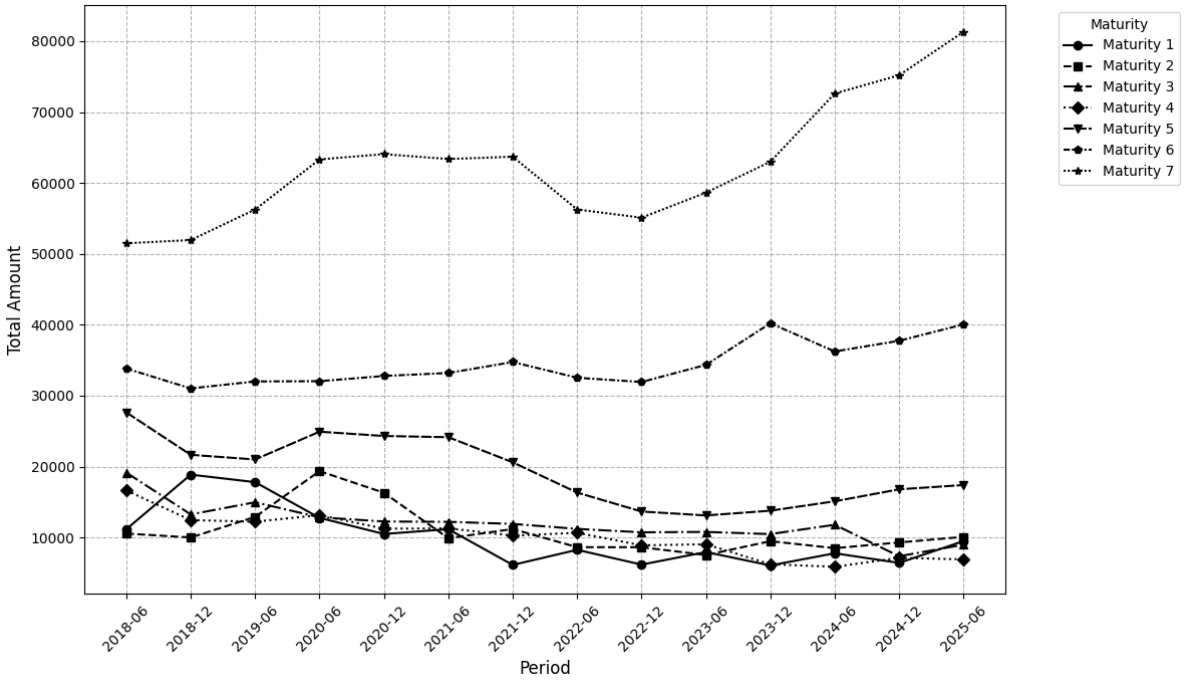


Figure 16: French Bond Holdings by Maturity

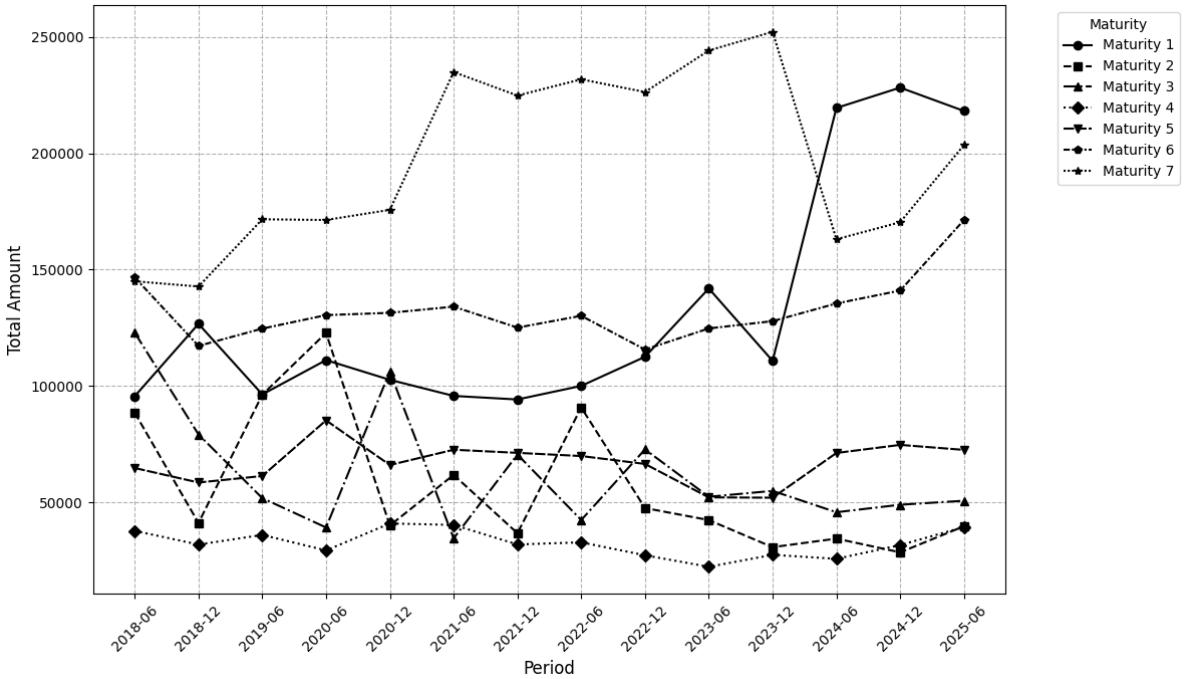


Figure 17: German Bond Holdings by Maturity

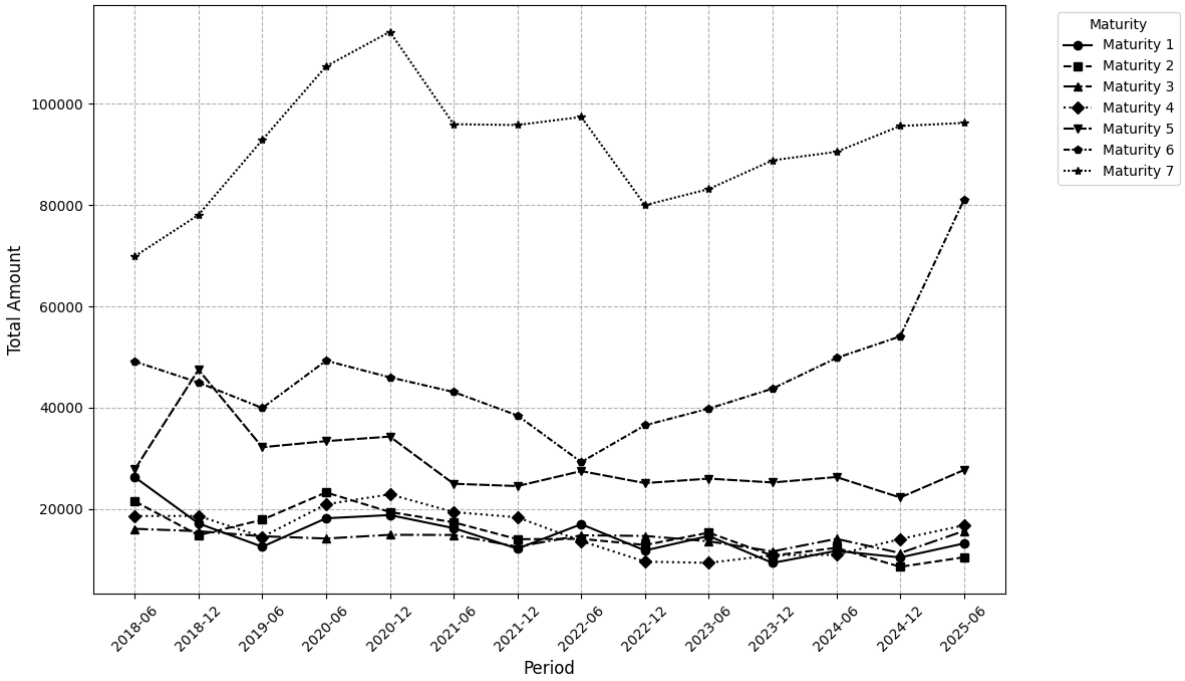


Figure 18: Greek Bond Holdings by Maturity

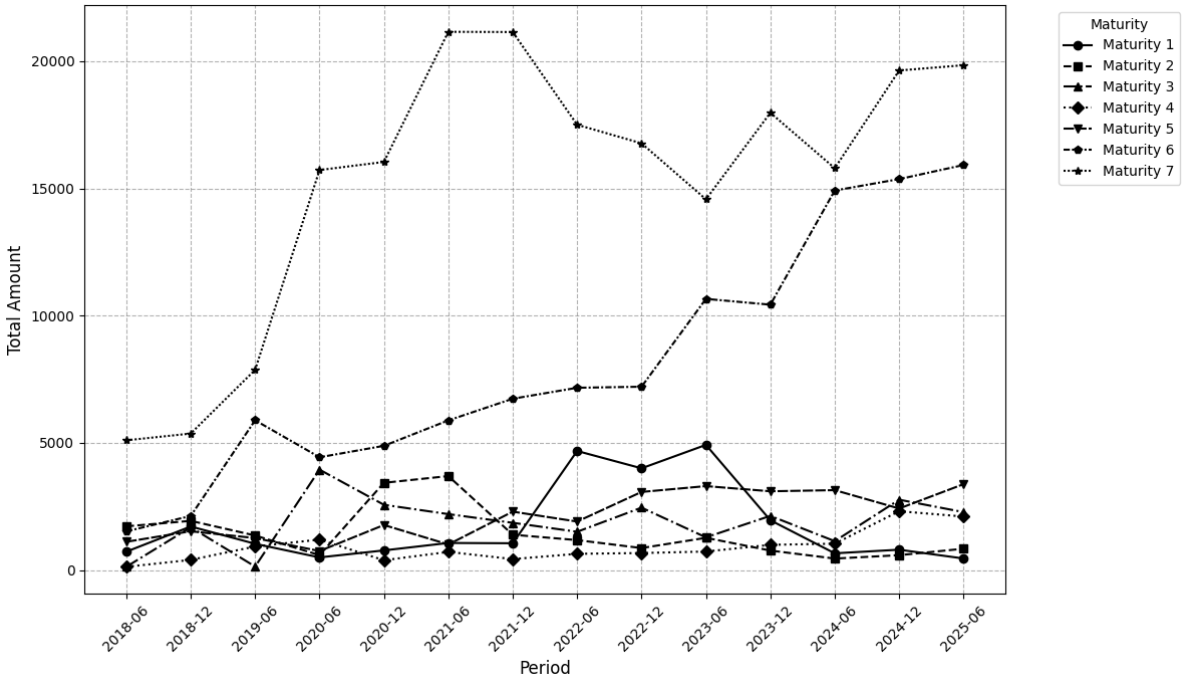


Figure 19: Irish Bond Holdings by Maturity

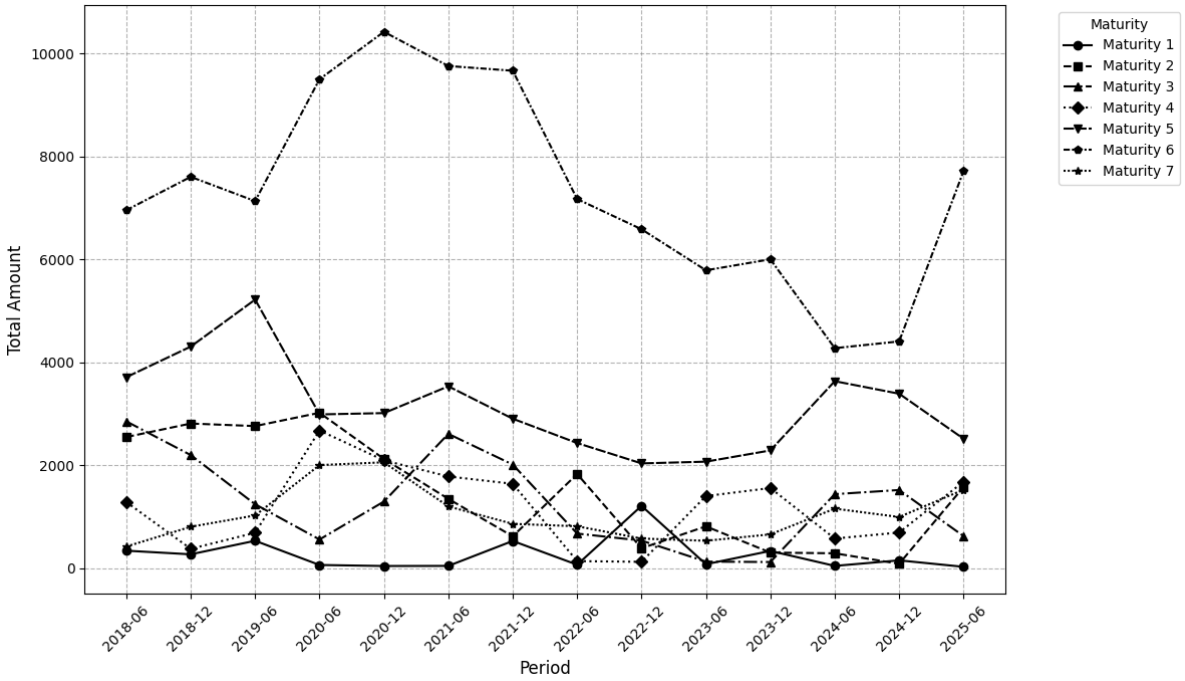


Figure 20: Italian Bond Holdings by Maturity

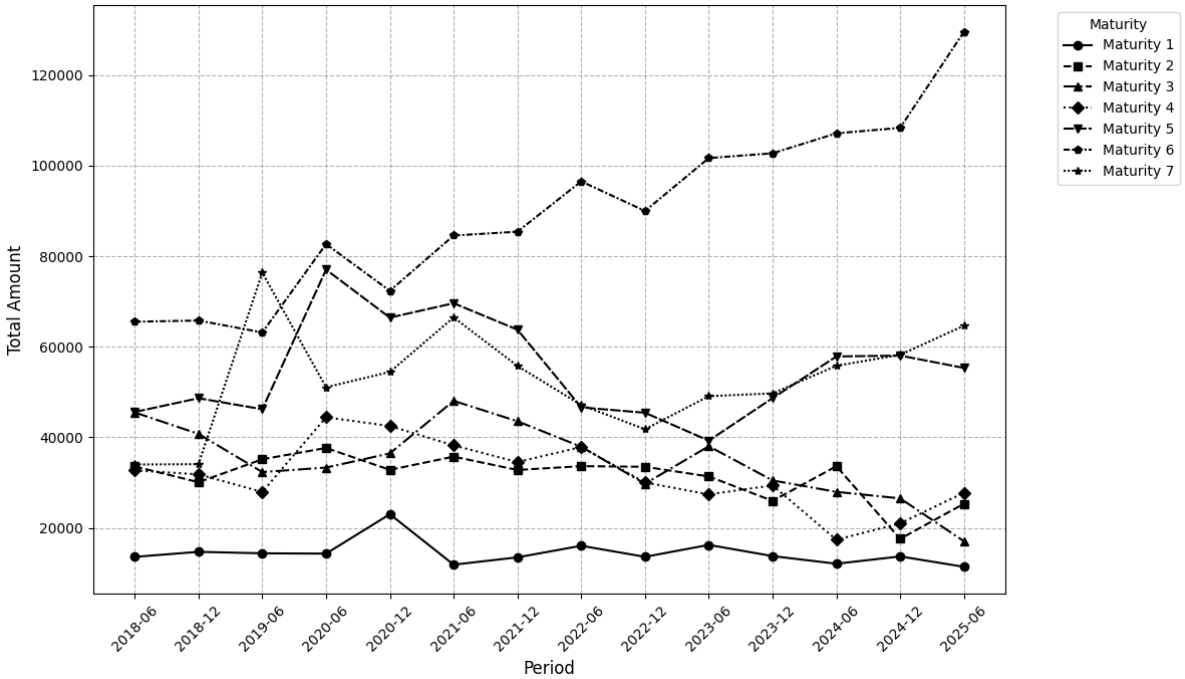


Figure 21: Portuguese Bond Holdings by Maturity

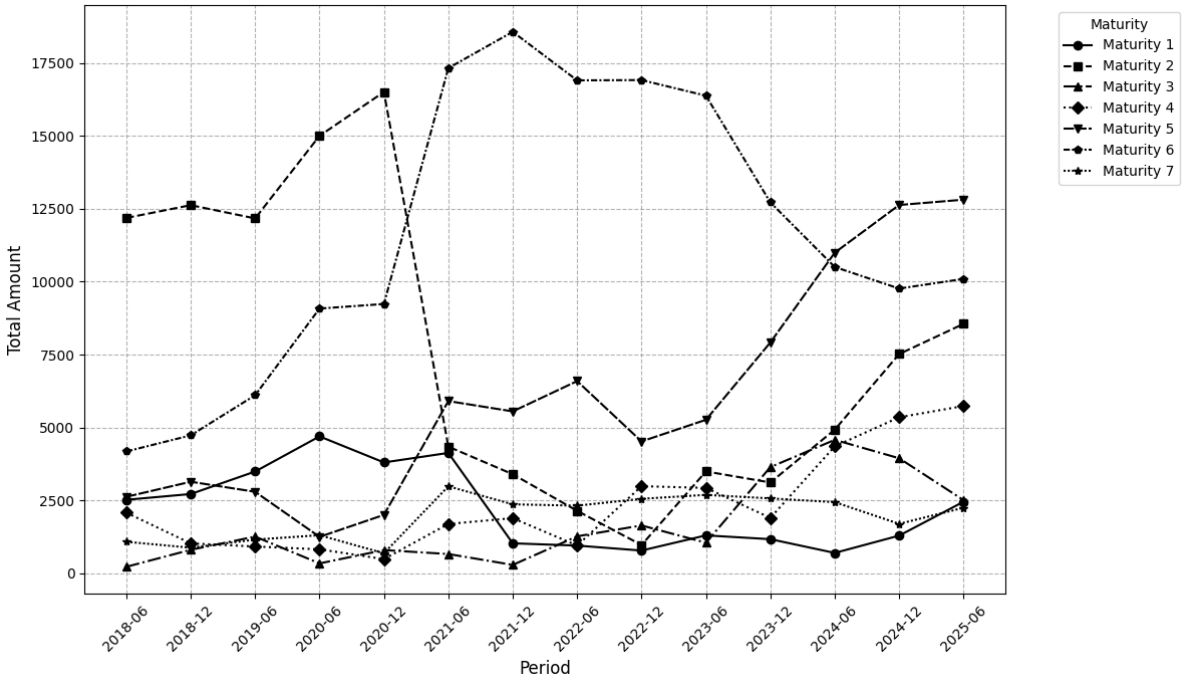
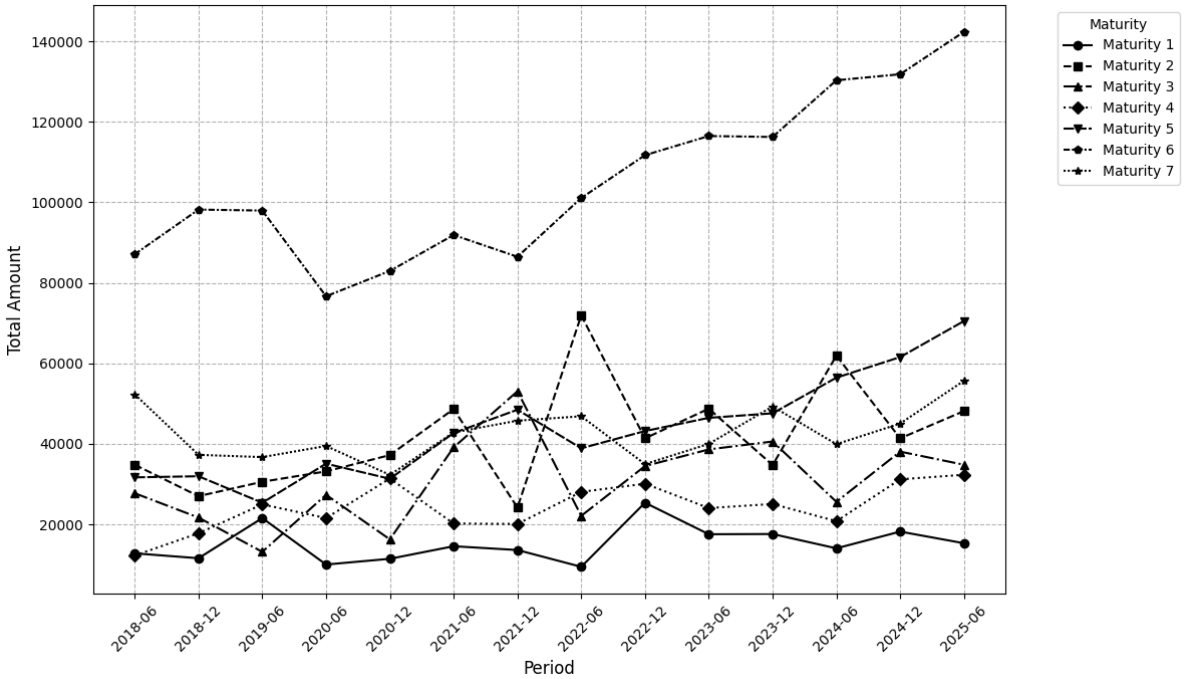


Figure 22: Spanish Bond Holdings by Maturity



B Mathematics

B.1 Instrument Extraction

Having obtained the residuals after rerunning the regression 14 times (once for each asset class identified in Ricco et al. (2024)) we construct a matrix $\widetilde{\widetilde{M}}_{T\times N}$ as follows.

$$\widetilde{\widetilde{M}} = \begin{pmatrix} \widetilde{\widetilde{ms}}_1^1 & \widetilde{\widetilde{ms}}_1^2 & \dots & \widetilde{\widetilde{ms}}_1^{14} \\ \widetilde{\widetilde{ms}}_2^1 & \widetilde{\widetilde{ms}}_2^2 & \dots & \widetilde{\widetilde{ms}}_2^{14} \\ \vdots & \vdots & \ddots & \vdots \\ \widetilde{\widetilde{ms}}_T^1 & \widetilde{\widetilde{ms}}_T^2 & \dots & \widetilde{\widetilde{ms}}_T^{14} \end{pmatrix}$$

Here the row index T refers to the index of the ECB meeting $t = 1, \dots, T$ and column index references, in order, OIS_{1m}, OIS_{3m}, OIS_{6m}, OIS_{1y}, OIS_{2y}, OIS_{5y}, OIS_{10}, GER-IT_{2y}, GER-IT_{5y}, GER-IT_{10y}, EUR/GBP, EUR/JPY, EUR/USD and STOXX50. Next, we note that the variance as well as the unit of measurement of each of these variables is inherently different and therefore, to continue our analysis, we construct a standardised matrix as below.

$$Y_{t,i} = \frac{\widetilde{\widetilde{ms}}_t^i - \overline{m}^i}{\hat{\sigma}^i}$$

Here, $\overline{m}^i$ represents the mean for variable i and $\hat{\sigma}^i$ represents the standard deviation. Having reassembled the matrix, each column now has zero mean and unit variance and the matrix Y represents the normalised responses of all 14 variables upon an ECB announcement day. We then construct the sample covariance matrix

$$\hat{\Sigma} = \frac{1}{T-1} Y'Y$$

which corresponds to the correlation matrix due to standardisation with all diagonal elements equalling 1 and off diagonal elements representing pairwise correlations (related proofs in **??**). We then continue with an eigendecomposition of $\hat{\Sigma}$ in the form

$$\hat{\Sigma} = V \Lambda_e V'$$

where V is the $N \times N$ matrix of eigenvectors arranged as columns and representing loading vectors, and $\Lambda_e = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_N)$ is the diagonal matrix of eigenvalues ordered in descending order, each representing the variance explained by the k-th principal component. We then conduct a test of weakness as described by Cragg and Donald (1993), where, in line with Ricco et al. (2024) we expect the first four of the eigenvalues to be significant. We then stack the significant loading vectors into the matrix $\hat{V}_4$ and construct the matrix of raw principal components

$$\hat{F} = Y\hat{V}_4$$

which is orthogonal by design. We now seek the 4×4 orthonormal matrix U such that $UU' = U'U = I_4$, $F = \hat{F}U$ and $\Lambda = \hat{V}_4 U$ satisfy the three key identification restrictions of Ricco et al. (2024).

Table 3: The Three Restrictions of Ricco et al. (2024)

Number	Description	Mathematical Notation	Rationale
1	Only F1 loads on OIS_{1m}	(B1)	Appendix C.1
2	Minimum pre-crisis variance of F3 and F4	(B2)	Appendix C.2
3	F4 has zero loading on OIS_{10y}	(B3)	Appendix C.3

Note that we parametrise U as $U = \exp(S)$ where

$$S = \begin{pmatrix} 0 & \theta_1 & \theta_2 & \theta_3 \\ -\theta_1 & 0 & \theta_4 & \theta_5 \\ -\theta_2 & -\theta_4 & 0 & \theta_6 \\ -\theta_3 & -\theta_5 & -\theta_6 & 0 \end{pmatrix}$$

and $\theta = (\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6)'$ fully parametrizes the rotation. Now we solve the following optimization problem.

$$\hat{\theta} = \arg\min_{\theta}[\text{Var}_{pre}(F_3(\theta)) + \text{Var}_{pre}(F_4(\theta)) + \rho(\Lambda^2_{j_{\text{OIS}_{1m}},2} + \Lambda^2_{j_{\text{OIS}_{1m}},3} + \Lambda^2_{j_{\text{OIS}_{1m}},4} + \Lambda^2_{j_{\text{OIS}_{10y}},4})]$$

Having obtained $\hat{\theta}$, we recover the rotated factors and loadings, normalise and rescale (further discussion on the mathematics in Appendix B.2) to obtain $\widetilde{F}^{k \times T}$ where each k represents the instruments for the 4 dimensions of ECB policy (normal monetary policy, forward guidance, QE/QT and country risk) that we are trying to instrument.

The

Having retrieved the residuals, we calculate the four factors identified by Ricco et al. (2024), where F1 corresponds to the factor of traditional monetary policy, F2 shows the effects of forward guidance, F3 represents QE/QT factors while F4 displays country risk. These factors are then fed into the Bayesian VAR specification of the authors

B.2 Marginal Likelihood Maximization

B.3 Gibbs Sampler

B.4 Miscellaneous Mathematics

B.4.1 Covariance Matrix Proofs

Theorem B.4.1.1. *The matrix $\hat{\Sigma}$ has all diagonal elements 1 and off-diagonal elements that represent correlations.*

Proof. Take any matrix $Y^{T\times K}$ of T time periods and K assets where $Y_{t,i}$ is defined as before and

$$Y = \begin{pmatrix} y_{1,1} & y_{1,2} & \cdots & y_{1,K} \\ y_{2,1} & y_{2,2} & \cdots & y_{2,K} \\ \vdots & \vdots & \ddots & \vdots \\ y_{T,1} & y_{T,2} & \cdots & y_{T,K} \end{pmatrix}$$

then

$$Y'Y = \begin{pmatrix} \sum_{t=1}^{T} y_{t,1}^2 & \sum_{t=1}^{T} y_{t,1}y_{t,2} & \cdots & \sum_{t=1}^{T} y_{t,1}y_{t,K} \\ \sum_{t=1}^{T} y_{t,2}y_{t,1} & \sum_{t=1}^{T} y_{t,2}^2 & \cdots & \sum_{t=1}^{T} y_{t,2}y_{t,K} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{t=1}^{T} y_{t,K}y_{t,1} & \sum_{t=1}^{T} y_{t,K}y_{t,2} & \cdots & \sum_{t=1}^{T} y_{t,K}^2 \end{pmatrix}$$

and with

$$\hat{\sigma}_{i,j} = \frac{1}{T-1}\sum_{t=1}^{T} y_{t,i}y_{t,j}$$

$\hat{\Sigma}$ becomes

$$\hat{\Sigma} = \frac{1}{T-1}Y'Y = \begin{pmatrix} \hat{\sigma}_{1,1} & \hat{\sigma}_{1,2} & \cdots & \hat{\sigma}_{1,K} \\ \hat{\sigma}_{2,1} & \hat{\sigma}_{2,2} & \cdots & \hat{\sigma}_{2,K} \\ \vdots & \vdots & \ddots & \vdots \\ \hat{\sigma}_{K,1} & \hat{\sigma}_{K,2} & \cdots & \hat{\sigma}_{K,K} \end{pmatrix}.$$

Now, for any series z with elements $z_1, z_2, z_3, \ldots, z_T$ with sample mean $\overline{z} = 0$, the sample variance formula becomes

$$\sigma^2 = \frac{1}{T-1}\sum_{t=1}^{T}(z_t - \overline{z})^2 = \frac{1}{T-1}\sum_{t=1}^{T} z_t^2.$$

Since all columns of Y have zero mean, we can set $z_t = y_{t,1}$ and observe that this is exactly $\hat{\sigma}_{ii}$. As the columns of Y were scaled to unit variance, $\hat{\sigma}_{ii} = 1$ for $\forall i \in \mathbb{R}$. Furthermore, covariance between two series z and w with elements $z_1, z_2, z_3, \ldots, z_T$ and

$w_1, w_2, w_3, \ldots, w_T$ with $\overline{z} = 0$ and $\overline{w} = 0$ is

$$\operatorname{cov}(z, w) = \frac{1}{T-1} \sum_{t=1}^{T} (z_t - \overline{z})(w_t - \overline{w}) = \frac{1}{T-1} \sum_{t=1}^{T} z_t w_t.$$

Now we set $z_t = y_{t,i}$ and $w_t = y_{t,j}$ where $i \neq j$ and note that the correlation formula

$$\rho_{ij} = \frac{\operatorname{cov}(y_{t,i}, y_{t,j})}{\sigma_i \cdot \sigma_j}$$

where $\sigma_i = \sigma_j = 1$ due to standardisation. This then implies that $\operatorname{cov}(y_{t,i}, y_{t,j}) = \rho_{ij}$ and $\hat{\Sigma}$ becomes

$$\hat{\Sigma} = \begin{pmatrix} 1 & \rho_{1,2} & \ldots & \rho_{1,K} \\ \rho_{2,1} & 1 & \ldots & \rho_{2,K} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{K,1} & \rho_{K,2} & \ldots & 1 \end{pmatrix}.$$

□

Theorem B.4.1.2. $\hat{\Sigma}$ *is symmetric and positive semi-definite.*

Proof. First, we note that only square matrices may be symmetric. When calculating $\hat{\Sigma}$ we multiply $Y'_{K\times T}$ with $Y_{T\times K}$ which implies $\hat{\Sigma}^{K\times K}$ by definition, which is square. For any matrix A to be symmetric the following must hold

$$A' = A.$$

Now we note that

$$\hat{\Sigma}' = \begin{pmatrix} 1 & \rho_{2,1} & \ldots & \rho_{K,1} \\ \rho_{1,2} & 1 & \ldots & \rho_{K,2} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{1,K} & \rho_{2,K} & \ldots & 1 \end{pmatrix}.$$

Now $\hat{\sigma}_{ii}$ is, by definition, equal to itself, additionally, since $\hat{\sigma}_{ij} = \rho_{ij}$ and $\rho_{1,2} = \rho_{2,1}$ by definition, $\hat{\sigma}_{ij} = \hat{\sigma}_{ji}$. For any matrix $A^{N\times N}$ to be positive semi-definite, the following must hold

$$a'Aa \geq 0$$

for $a \in \mathbb{R}^{N^+}$. Now we take the definition of $\hat{\Sigma}$ and note

$$a'\hat{\Sigma}a = \frac{1}{T-1} a'Y'Ya.$$

Defining $z = Ya$ we note that the t-th entry is $\sum_{i=1}^{T} y_{t,i} a_i$ we obtain

$$a'\hat{\Sigma}a = \frac{1}{T-1} z'z = \frac{1}{T-1} \sum_{i=1}^{T} z_t^2 = \frac{1}{T-1} \sum_{i=1}^{T} \left(\sum_{i=1}^{K} y_{t,i} a_i \right)^2$$

where $z_t^2 \geq 0$ for $\forall t \in \mathbb{R}^+$, thus

$$a'\hat{\Sigma}a \geq 0 \text{ for } \forall a \in \mathbb{R}^{\mathrm{K}^+}.$$

□

Restriction 1

$$\Lambda_{j_{\mathrm{OIS}_{1m}},2} = 0\,; \quad \Lambda_{j_{\mathrm{OIS}_{1m}},3} = 0\,; \quad \Lambda_{j_{\mathrm{OIS}_{1m}},4} = 0 \tag{B1}$$

Where $\Lambda_{i,k}$ is the loading of asset i and factor k.

Restriction 2

$$\mathrm{Var}_{pre}(F_k) = \frac{1}{|\mathcal{T}_{pre} - 1|} \sum_{t \in \mathcal{T}_{pre}} \left(F_{t,k} - \overline{F}_l^{pre} \right)^2 \tag{B2}$$

Where $\mathcal{T}_{pre} = \{t : t < \text{August 2008}\}$ is the set of meeting indices pre-crisis and $\overline{F}_k^{pre}$ is the sample mean of factor k before the crisis.

Restriction 3

$$\Lambda_{j_{\mathrm{OIS}_{10y}},4} = 0 \tag{B3}$$

C Extensive Arguments

C.1 Restriction 1

C.2 Restriction 2

C.3 Restriction 3